

To the editors of Nature Health

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Dear editors,

We are pleased to submit our manuscript entitled **“COVID-19 disrupted patterns of cause-specific mortality in Switzerland”** for consideration in *Nature Health*. Using high-quality national mortality data and a Bayesian multivariate model, we quantify cause-specific excess mortality during the COVID-19 pandemic and reveal how mortality from different causes is interdependent.

We use data from 2011–2019 to model expected mortality and assess cause-specific excess deaths during 2020–2021 across nine aggregated cause groups. The model accounts for long-term trends, seasonality, population changes, and cross-cause correlations, allowing us to identify how the COVID-19 pandemic disrupted historically synchronized mortality patterns. We show that non-COVID respiratory (–25%) and mental/neurological (–12%) deaths declined among individuals aged ≥80, largely offset by COVID-19 deaths. We also show that cardiovascular and mental/neurological excesses correlated with COVID-19 and other respiratory excesses, reflecting strong inferred pre-pandemic cross-cause correlations.

By leveraging the unique situation created by the COVID-19 pandemic, which temporarily decoupled some cause-specific mortality trends, this study identifies the mechanisms driving shifts in excess mortality across causes during the pandemic. Specifically, it highlights three overlapping mechanisms: reduced circulation of non-COVID-19 respiratory pathogens, unrecognized COVID-19 deaths (especially cardiovascular), and mortality displacement. Beyond explaining the cause-specific mortality patterns observed during the pandemic, this also provides insight into the nature of historical correlations between certain causes of death, such as those between cardiovascular and infectious diseases.

We believe these findings have clear implications for health policy and mortality surveillance, pointing to a potential underestimation of the true burden of respiratory pathogens and underscoring the value of multiple-cause-of-death data to better characterize these mechanisms. This work sits at the intersection of infectious disease, disease epidemiology, and health policy, which are key areas of interest for *Nature Health*.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Dr. Anthony Hauser on behalf of all authors